

UNIT-8 WEB PROGRAMMING USING PYTHON

API - Application Programming Interface

What is an API?

An application programming interface is a way for two or more computer programs to communicate with each other. It is a type of software interface, offering a service to other pieces of software. A document or standard that describes how to build or use such a connection or interface is called an API specification.

The OpenWeatherMap API <https://openweathermap.org/api>

The OpenWeatherMap is a service that provides weather data, including current weather data, forecasts, and historical data to the developers of web services and mobile applications.

It provides an API with JSON, XML, and HTML endpoints and a limited free usage tier. Making more than 60 calls per minute requires a paid subscription starting at USD 40 per month. Access to historical data requires a subscription starting at 150 USD per month. Users can request current weather information, extended forecasts, and graphical maps (showing cloud cover, wind speed, pressure, and precipitation).

To use this current weather data API, one must need the API key.

Pricing and API Key

<https://openweathermap.org/price>

API Error Code Meaning

<https://openweathermap.org/faq#error401>

Geocoding API

<https://openweathermap.org/api/geocoding-api>

Geocoding API is a simple tool that we have developed to ease the search for locations while working with geographic names and coordinates.

Coordinates by location name

API Call

```
In [3]: #http://api.openweathermap.org/geo/1.0/direct?q={city name},{state code},{country code}
```

```
In [1]: import requests, json
```

```
def get_coordinates():
    api_key = input('Enter your API key: ')
    city = input('Enter the city name: ')
    request_url = "http://api.openweathermap.org/geo/1.0/direct?q=" + city + "&appid="
    response = requests.get(request_url)
    coordinates = response.json()
    print(coordinates)
    latitude = coordinates[0]['lat']
    longitude = coordinates[0]['lon']
    return (latitude, longitude)
```

```
lat, long = get_coordinates()
```

Enter your API key: 9e5a8fa0d6c89027bfd17698d38e35b5

Enter the city name: Ahmedabad

```
[{'name': 'Ahmedabad', 'local_names': {'hi': 'अहमदाबाद', 'mr': 'अहमदाबाद', 'pa': 'ਅਹਿਮਦਾਬਾਦ', 'eo': 'Ahmadabado', 'te': 'అహ్మదాబాద్', 'ja': 'アフマダーバード', 'ar': 'أحمد آباد', 'ml': 'അഹ്മദാബാദ്', 'kn': 'ಅಹ್ಮದಾಬಾದ್', 'ta': 'அகமதாபாத்', 'pl': 'Ahmadabad', 'ur': 'احمد آباد', 'or': 'ଅହମଦାବାଦ', 'ascii': 'Ahmedabad', 'de': 'Ahmedabad', 'en': 'Ahmedabad', 'zh': '艾哈迈达巴德', 'uk': 'Ахмедабад', 'cs': 'Ahmadábád', 'oc': 'Ahmadabad', 'he': 'אחמדאבאד', 'gu': 'અમદાવાદ', 'ru': 'Ахмадабад', 'feature_name': 'Ahmedabad'}, 'lat': 23.0216238, 'lon': 72.5797068, 'country': 'IN', 'state': 'Gujarat'}]
```

```
In [2]: print("Latitude & Longitude of Ahmedabad from Geocode API")
print("-----")
print("Latitude:", lat)
print("Longitude:", long)
```

Lattitude & Longitude of Ahmedabad from Geocode API

Lattitude: 23.0216238

Longitude: 72.5797068

Current Weather Data API

<https://openweathermap.org/current>

Used to obtain current weather details of any city.

Access current weather data for any location on Earth including over 200,000 cities! We collect and process weather data from different sources such as global and local weather models, satellites, radars and a vast network of weather stations. Data is available in JSON, XML, or HTML format.

Call current weather data

API Call

```
In [11]: #https://api.openweathermap.org/data/2.5/weather?lat={lat}&lon={lon}&appid={API key}
```

```
In [14]: # Here we are taking lat long from Geocoding API
```

```
In [3]: import requests, json
def display_current_weather():

    api_key = input('Enter your API key: ')
    request_url = "https://api.openweathermap.org/data/2.5/weather?lat=" + str(lat) +
    response = requests.get(request_url)
    current_weather = response.json()
```

```

print(json.dumps(current_weather, indent = 5))
print()
description = current_weather['weather'][0]['description']
temperature = current_weather['main']['temp']
pressure = current_weather['main']['pressure']
humidity = current_weather['main']['humidity']
wind_speed = current_weather['wind']['speed']
visibility = current_weather['visibility']

print('Current Weather')
print("-----")
print('Temperature', temperature)
print('Pressure', pressure)
print('Humidity', humidity)
print('Wind Speed', wind_speed)
print('Visibility', visibility)

display_current_weather()

```

Enter your API key: 9e5a8fa0d6c89027bfd17698d38e35b5

```

{
  "coord": {
    "lon": 72.5797,
    "lat": 23.0216
  },
  "weather": [
    {
      "id": 501,
      "main": "Rain",
      "description": "moderate rain",
      "icon": "10d"
    }
  ],
  "base": "stations",
  "main": {
    "temp": 300.19,
    "feels_like": 304.44,
    "temp_min": 300.19,
    "temp_max": 300.19,
    "pressure": 998,
    "humidity": 94,
    "sea_level": 998,
    "grnd_level": 993
  },
  "visibility": 2000,
  "wind": {
    "speed": 4.12,
    "deg": 210
  },
  "rain": {
    "1h": 1.76
  },
  "clouds": {
    "all": 100
  },
  "dt": 1722226613,
  "sys": {
    "type": 1,
    "id": 9049,
    "country": "IN",
    "sunrise": 1722213547,
    "sunset": 1722261178
  },
  "timezone": 19800,

```

```

    "id": 1279233,
    "name": "Ahmedabad",
    "cod": 200
}

```

Current Weather

```

-----
Temperature 300.19
Pressure 998
Humidity 94
Wind Speed 4.12
Visibility 2000

```

Air Pollution API

<https://openweathermap.org/api/air-pollution>

Used to obtain current air pollution data, air pollution forecast and also past air pollution data of any city.

Air Pollution API provides current, forecast and historical air pollution data for any coordinates on the globe.

Besides basic Air Quality Index, the API returns data about polluting gases, such as Carbon monoxide (CO), Nitrogen monoxide (NO), Nitrogen dioxide (NO₂), Ozone (O₃), Sulphur dioxide (SO₂), Ammonia (NH₃), and particulates (PM_{2.5} and PM₁₀).

Here is a description of Air Quality Index levels:

Qualitative name	Index	Pollutant concentration in µg/m ³					
		SO ₂	NO ₂	PM ₁₀	PM _{2.5}	O ₃	CO
Good	1	[0; 20)	[0; 40)	[0; 20)	[0; 10)	[0; 60)	[0; 4400)
Fair	2	[20; 80)	[40; 70)	[20; 50)	[10; 25)	[60; 100)	[4400; 9400)
Moderate	3	[80; 250)	[70; 150)	[50; 100)	[25; 50)	[100; 140)	[9400-12400)
Poor	4	[250; 350)	[150; 200)	[100; 200)	[50; 75)	[140; 180)	[12400; 15400)
Very Poor	5	≥350	≥200	≥200	≥75	≥180	≥15400

Current Air Pollution Data

API Call

In [17]: `#http://api.openweathermap.org/data/2.5/air_pollution?lat={lat}&lon={lon}&appid={API key}`

```
In [4]: import requests,json

quality_indices = {1:'Good', 2:'Fair', 3:'Moderate', 4:'Poor', 5:'Very Poor'}

def display_current_air_pollution():

    api_key = input('Enter your API key: ')
    request_url = "http://api.openweathermap.org/data/2.5/air_pollution?lat="+str(lat)
    response = requests.get(request_url)
    current_pollution = response.json()
    print(json.dumps(current_pollution, indent = 5))

    index = current_pollution['list'][0]['main']['aqi']
    print('Current Air Quality: ', quality_indices[index])

display_current_air_pollution()
```

Enter your API key: 9e5a8fa0d6c89027bfd17698d38e35b5

```
{
  "coord": {
    "lon": 72.5797,
    "lat": 23.0216
  },
  "list": [
    {
      "main": {
        "aqi": 2
      },
      "components": {
        "co": 270.37,
        "no": 0.27,
        "no2": 7.11,
        "o3": 16.81,
        "so2": 2.27,
        "pm2_5": 12.08,
        "pm10": 14.88,
        "nh3": 1.2
      },
      "dt": 1722226627
    }
  ]
}
Current Air Quality: Fair
```

5 Day Weather Forecast API

<https://openweathermap.org/forecast5>

5 day forecast is available at any location on the globe. It includes weather forecast data with 3-hour step. Forecast is available in JSON or XML format.

Here we are generating 5 Day weather forecast for Ahmedabad.

```
In [5]: import requests,json

def five_day_forecast():
    api_key = input('Enter your API key: ')
    request_url = "http://api.openweathermap.org/data/2.5/forecast?lat=" + str(lat) +
    response = requests.get(request_url)
    five_day = response.json()
```

```
print(json.dumps(five_day, indent = 5))
return five_day
```

```
In [6]: five_day = five_day_forecast()
```

Enter your API key: 9e5a8fa0d6c89027bfd17698d38e35b5

```
{
  "cod": "200",
  "message": 0,
  "cnt": 40,
  "list": [
    {
      "dt": 1722232800,
      "main": {
        "temp": 300.81,
        "feels_like": 305.54,
        "temp_min": 300.81,
        "temp_max": 302.05,
        "pressure": 997,
        "sea_level": 997,
        "grnd_level": 992,
        "humidity": 88,
        "temp_kf": -1.24
      },
      "weather": [
        {
          "id": 500,
          "main": "Rain",
          "description": "light rain",
          "icon": "10d"
        }
      ],
      "clouds": {
        "all": 100
      },
      "wind": {
        "speed": 4.66,
        "deg": 254,
        "gust": 6.02
      },
      "visibility": 10000,
      "pop": 1,
      "rain": {
        "3h": 1.2
      },
      "sys": {
        "pod": "d"
      },
      "dt_txt": "2024-07-29 06:00:00"
    },
    {
      "dt": 1722243600,
      "main": {
        "temp": 301.02,
        "feels_like": 305.53,
        "temp_min": 301.02,
        "temp_max": 301.43,
        "pressure": 996,
        "sea_level": 996,
        "grnd_level": 991,
        "humidity": 84,
        "temp_kf": -0.41
      },
      "weather": [
```

```

        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 5.4,
        "deg": 233,
        "gust": 6.97
    },
    "visibility": 10000,
    "pop": 1,
    "rain": {
        "3h": 1.23
    },
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-07-29 09:00:00"
},
{
    "dt": 1722254400,
    "main": {
        "temp": 301.01,
        "feels_like": 305.2,
        "temp_min": 301.01,
        "temp_max": 301.01,
        "pressure": 996,
        "sea_level": 996,
        "grnd_level": 991,
        "humidity": 82,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 5.28,
        "deg": 210,
        "gust": 6.8
    },
    "visibility": 10000,
    "pop": 1,
    "rain": {
        "3h": 1.76
    },
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-07-29 12:00:00"
},
{

```

```

"dt": 1722265200,
"main": {
  "temp": 299.86,
  "feels_like": 302.99,
  "temp_min": 299.86,
  "temp_max": 299.86,
  "pressure": 997,
  "sea_level": 997,
  "grnd_level": 992,
  "humidity": 89,
  "temp_kf": 0
},
"weather": [
  {
    "id": 500,
    "main": "Rain",
    "description": "light rain",
    "icon": "10n"
  }
],
"clouds": {
  "all": 100
},
"wind": {
  "speed": 4.33,
  "deg": 204,
  "gust": 7.39
},
"visibility": 5375,
"pop": 1,
"rain": {
  "3h": 2.92
},
"sys": {
  "pod": "n"
},
"dt_txt": "2024-07-29 15:00:00"
},
{
  "dt": 1722276000,
  "main": {
    "temp": 299.93,
    "feels_like": 303.09,
    "temp_min": 299.93,
    "temp_max": 299.93,
    "pressure": 999,
    "sea_level": 999,
    "grnd_level": 993,
    "humidity": 88,
    "temp_kf": 0
  },
  "weather": [
    {
      "id": 500,
      "main": "Rain",
      "description": "light rain",
      "icon": "10n"
    }
  ],
  "clouds": {
    "all": 100
  },
  "wind": {
    "speed": 4.27,
    "deg": 210,

```



```

        "gust": 8.02
      },
      "visibility": 10000,
      "pop": 1,
      "rain": {
        "3h": 1.16
      },
      "sys": {
        "pod": "n"
      },
      "dt_txt": "2024-07-29 18:00:00"
    },
    {
      "dt": 1722286800,
      "main": {
        "temp": 299.83,
        "feels_like": 302.91,
        "temp_min": 299.83,
        "temp_max": 299.83,
        "pressure": 998,
        "sea_level": 998,
        "grnd_level": 993,
        "humidity": 89,
        "temp_kf": 0
      },
      "weather": [
        {
          "id": 500,
          "main": "Rain",
          "description": "light rain",
          "icon": "10n"
        }
      ],
      "clouds": {
        "all": 100
      },
      "wind": {
        "speed": 3.49,
        "deg": 238,
        "gust": 6.42
      },
      "visibility": 8511,
      "pop": 1,
      "rain": {
        "3h": 1.56
      },
      "sys": {
        "pod": "n"
      },
      "dt_txt": "2024-07-29 21:00:00"
    },
    {
      "dt": 1722297600,
      "main": {
        "temp": 299.79,
        "feels_like": 299.79,
        "temp_min": 299.79,
        "temp_max": 299.79,
        "pressure": 999,
        "sea_level": 999,
        "grnd_level": 993,
        "humidity": 87,
        "temp_kf": 0
      },
      "weather": [

```

```

        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 3.63,
        "deg": 224,
        "gust": 5.58
    },
    "visibility": 10000,
    "pop": 1,
    "rain": {
        "3h": 0.9
    },
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2024-07-30 00:00:00"
},
{
    "dt": 1722308400,
    "main": {
        "temp": 299.71,
        "feels_like": 299.71,
        "temp_min": 299.71,
        "temp_max": 299.71,
        "pressure": 1001,
        "sea_level": 1001,
        "grnd_level": 995,
        "humidity": 87,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 3.45,
        "deg": 224,
        "gust": 6.18
    },
    "visibility": 10000,
    "pop": 0.51,
    "rain": {
        "3h": 0.46
    },
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-07-30 03:00:00"
},
{

```

```

"dt": 1722319200,
"main": {
  "temp": 301.42,
  "feels_like": 304.95,
  "temp_min": 301.42,
  "temp_max": 301.42,
  "pressure": 1001,
  "sea_level": 1001,
  "grnd_level": 996,
  "humidity": 74,
  "temp_kf": 0
},
"weather": [
  {
    "id": 804,
    "main": "Clouds",
    "description": "overcast clouds",
    "icon": "04d"
  }
],
"clouds": {
  "all": 100
},
"wind": {
  "speed": 3.5,
  "deg": 234,
  "gust": 5.38
},
"visibility": 10000,
"pop": 0.44,
"sys": {
  "pod": "d"
},
"dt_txt": "2024-07-30 06:00:00"
},
{
  "dt": 1722330000,
  "main": {
    "temp": 304.17,
    "feels_like": 308.29,
    "temp_min": 304.17,
    "temp_max": 304.17,
    "pressure": 999,
    "sea_level": 999,
    "grnd_level": 994,
    "humidity": 61,
    "temp_kf": 0
  },
  "weather": [
    {
      "id": 804,
      "main": "Clouds",
      "description": "overcast clouds",
      "icon": "04d"
    }
  ],
  "clouds": {
    "all": 100
  },
  "wind": {
    "speed": 2.51,
    "deg": 255,
    "gust": 4.8
  },
  "visibility": 10000,

```

```

    "pop": 0,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2024-07-30 09:00:00"
  },
  {
    "dt": 1722340800,
    "main": {
      "temp": 305.25,
      "feels_like": 309.86,
      "temp_min": 305.25,
      "temp_max": 305.25,
      "pressure": 999,
      "sea_level": 999,
      "grnd_level": 994,
      "humidity": 58,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 804,
        "main": "Clouds",
        "description": "overcast clouds",
        "icon": "04d"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 1.93,
      "deg": 331,
      "gust": 2.69
    },
    "visibility": 10000,
    "pop": 0.02,
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2024-07-30 12:00:00"
  },
  {
    "dt": 1722351600,
    "main": {
      "temp": 302.72,
      "feels_like": 307.61,
      "temp_min": 302.72,
      "temp_max": 302.72,
      "pressure": 1000,
      "sea_level": 1000,
      "grnd_level": 995,
      "humidity": 72,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 500,
        "main": "Rain",
        "description": "light rain",
        "icon": "10n"
      }
    ],
    "clouds": {
      "all": 97
    }
  }

```

```

    },
    "wind": {
      "speed": 3.49,
      "deg": 180,
      "gust": 5.91
    },
    "visibility": 10000,
    "pop": 0.59,
    "rain": {
      "3h": 0.69
    },
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-07-30 15:00:00"
  },
  {
    "dt": 1722362400,
    "main": {
      "temp": 302.16,
      "feels_like": 306.67,
      "temp_min": 302.16,
      "temp_max": 302.16,
      "pressure": 1002,
      "sea_level": 1002,
      "grnd_level": 997,
      "humidity": 74,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 804,
        "main": "Clouds",
        "description": "overcast clouds",
        "icon": "04n"
      }
    ],
    "clouds": {
      "all": 93
    },
    "wind": {
      "speed": 3.93,
      "deg": 224,
      "gust": 8.39
    },
    "visibility": 10000,
    "pop": 0.51,
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-07-30 18:00:00"
  },
  {
    "dt": 1722373200,
    "main": {
      "temp": 300.75,
      "feels_like": 304.4,
      "temp_min": 300.75,
      "temp_max": 300.75,
      "pressure": 1002,
      "sea_level": 1002,
      "grnd_level": 996,
      "humidity": 81,
      "temp_kf": 0
    },

```

```

    "weather": [
      {
        "id": 500,
        "main": "Rain",
        "description": "light rain",
        "icon": "10n"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 4.56,
      "deg": 228,
      "gust": 8.5
    },
    "visibility": 10000,
    "pop": 0.81,
    "rain": {
      "3h": 0.6
    },
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-07-30 21:00:00"
  },
  {
    "dt": 1722384000,
    "main": {
      "temp": 300.14,
      "feels_like": 303.25,
      "temp_min": 300.14,
      "temp_max": 300.14,
      "pressure": 1002,
      "sea_level": 1002,
      "grnd_level": 997,
      "humidity": 84,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 804,
        "main": "Clouds",
        "description": "overcast clouds",
        "icon": "04n"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 4.05,
      "deg": 216,
      "gust": 7.04
    },
    "visibility": 10000,
    "pop": 0.45,
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-07-31 00:00:00"
  },
  {
    "dt": 1722394800,
    "main": {

```

```

        "temp": 301.75,
        "feels_like": 305.88,
        "temp_min": 301.75,
        "temp_max": 301.75,
        "pressure": 1003,
        "sea_level": 1003,
        "grnd_level": 998,
        "humidity": 75,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 804,
            "main": "Clouds",
            "description": "overcast clouds",
            "icon": "04d"
        }
    ],
    "clouds": {
        "all": 99
    },
    "wind": {
        "speed": 4.47,
        "deg": 236,
        "gust": 5.92
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-07-31 03:00:00"
},
{
    "dt": 1722405600,
    "main": {
        "temp": 305.97,
        "feels_like": 310.25,
        "temp_min": 305.97,
        "temp_max": 305.97,
        "pressure": 1003,
        "sea_level": 1003,
        "grnd_level": 998,
        "humidity": 54,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
        }
    ],
    "clouds": {
        "all": 95
    },
    "wind": {
        "speed": 5.46,
        "deg": 250,
        "gust": 6.41
    },
    "visibility": 10000,
    "pop": 0.2,
    "rain": {

```

```

        "3h": 0.19
    },
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-07-31 06:00:00"
},
{
    "dt": 1722416400,
    "main": {
        "temp": 306.53,
        "feels_like": 309.92,
        "temp_min": 306.53,
        "temp_max": 306.53,
        "pressure": 1002,
        "sea_level": 1002,
        "grnd_level": 996,
        "humidity": 49,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
        }
    ],
    "clouds": {
        "all": 88
    },
    "wind": {
        "speed": 4.86,
        "deg": 256,
        "gust": 6.44
    },
    "visibility": 10000,
    "pop": 0.2,
    "rain": {
        "3h": 0.2
    },
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-07-31 09:00:00"
},
{
    "dt": 1722427200,
    "main": {
        "temp": 305.32,
        "feels_like": 308.87,
        "temp_min": 305.32,
        "temp_max": 305.32,
        "pressure": 1000,
        "sea_level": 1000,
        "grnd_level": 995,
        "humidity": 54,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 804,
            "main": "Clouds",
            "description": "overcast clouds",
            "icon": "04d"
        }
    ]
}

```



```

    }
  ],
  "clouds": {
    "all": 94
  },
  "wind": {
    "speed": 3.42,
    "deg": 265,
    "gust": 6.44
  },
  "visibility": 10000,
  "pop": 0,
  "sys": {
    "pod": "d"
  },
  "dt_txt": "2024-07-31 12:00:00"
},
{
  "dt": 1722438000,
  "main": {
    "temp": 304.06,
    "feels_like": 308.05,
    "temp_min": 304.06,
    "temp_max": 304.06,
    "pressure": 1002,
    "sea_level": 1002,
    "grnd_level": 996,
    "humidity": 61,
    "temp_kf": 0
  },
  "weather": [
    {
      "id": 804,
      "main": "Clouds",
      "description": "overcast clouds",
      "icon": "04n"
    }
  ],
  "clouds": {
    "all": 100
  },
  "wind": {
    "speed": 5.73,
    "deg": 239,
    "gust": 9.92
  },
  "visibility": 10000,
  "pop": 0,
  "sys": {
    "pod": "n"
  },
  "dt_txt": "2024-07-31 15:00:00"
},
{
  "dt": 1722448800,
  "main": {
    "temp": 302.43,
    "feels_like": 306.5,
    "temp_min": 302.43,
    "temp_max": 302.43,
    "pressure": 1003,
    "sea_level": 1003,
    "grnd_level": 997,
    "humidity": 70,
    "temp_kf": 0
  },

```

```

    },
    "weather": [
      {
        "id": 804,
        "main": "Clouds",
        "description": "overcast clouds",
        "icon": "04n"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 4.99,
      "deg": 254,
      "gust": 9.5
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-07-31 18:00:00"
  },
  {
    "dt": 1722459600,
    "main": {
      "temp": 301.2,
      "feels_like": 304.91,
      "temp_min": 301.2,
      "temp_max": 301.2,
      "pressure": 1001,
      "sea_level": 1001,
      "grnd_level": 996,
      "humidity": 77,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 804,
        "main": "Clouds",
        "description": "overcast clouds",
        "icon": "04n"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 5.66,
      "deg": 244,
      "gust": 10.25
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-07-31 21:00:00"
  },
  {
    "dt": 1722470400,
    "main": {
      "temp": 300.57,
      "feels_like": 303.85,

```

```

        "temp_min": 300.57,
        "temp_max": 300.57,
        "pressure": 1001,
        "sea_level": 1001,
        "grnd_level": 996,
        "humidity": 80,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 804,
            "main": "Clouds",
            "description": "overcast clouds",
            "icon": "04n"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 4.99,
        "deg": 239,
        "gust": 9.84
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2024-08-01 00:00:00"
},
{
    "dt": 1722481200,
    "main": {
        "temp": 301.63,
        "feels_like": 305.26,
        "temp_min": 301.63,
        "temp_max": 301.63,
        "pressure": 1003,
        "sea_level": 1003,
        "grnd_level": 997,
        "humidity": 73,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 804,
            "main": "Clouds",
            "description": "overcast clouds",
            "icon": "04d"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 6.26,
        "deg": 238,
        "gust": 9.28
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
        "pod": "d"
    },

```

```

    "dt_txt": "2024-08-01 03:00:00"
  },
  {
    "dt": 1722492000,
    "main": {
      "temp": 303.74,
      "feels_like": 307.61,
      "temp_min": 303.74,
      "temp_max": 303.74,
      "pressure": 1002,
      "sea_level": 1002,
      "grnd_level": 997,
      "humidity": 62,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 500,
        "main": "Rain",
        "description": "light rain",
        "icon": "10d"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 7.11,
      "deg": 239,
      "gust": 9.51
    },
    "visibility": 10000,
    "pop": 0.2,
    "rain": {
      "3h": 0.21
    },
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2024-08-01 06:00:00"
  },
  {
    "dt": 1722502800,
    "main": {
      "temp": 305.85,
      "feels_like": 309.99,
      "temp_min": 305.85,
      "temp_max": 305.85,
      "pressure": 1000,
      "sea_level": 1000,
      "grnd_level": 994,
      "humidity": 54,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 500,
        "main": "Rain",
        "description": "light rain",
        "icon": "10d"
      }
    ],
    "clouds": {
      "all": 99
    },
  },

```

```

        "wind": {
            "speed": 6.63,
            "deg": 229,
            "gust": 8.92
        },
        "visibility": 10000,
        "pop": 1,
        "rain": {
            "3h": 0.46
        },
        "sys": {
            "pod": "d"
        },
        "dt_txt": "2024-08-01 09:00:00"
    },
    {
        "dt": 1722513600,
        "main": {
            "temp": 304.09,
            "feels_like": 308.63,
            "temp_min": 304.09,
            "temp_max": 304.09,
            "pressure": 999,
            "sea_level": 999,
            "grnd_level": 994,
            "humidity": 63,
            "temp_kf": 0
        },
        "weather": [
            {
                "id": 501,
                "main": "Rain",
                "description": "moderate rain",
                "icon": "10d"
            }
        ],
        "clouds": {
            "all": 100
        },
        "wind": {
            "speed": 5.64,
            "deg": 235,
            "gust": 9.2
        },
        "visibility": 10000,
        "pop": 1,
        "rain": {
            "3h": 4.1
        },
        "sys": {
            "pod": "d"
        },
        "dt_txt": "2024-08-01 12:00:00"
    },
    {
        "dt": 1722524400,
        "main": {
            "temp": 302.04,
            "feels_like": 306.19,
            "temp_min": 302.04,
            "temp_max": 302.04,
            "pressure": 1001,
            "sea_level": 1001,
            "grnd_level": 996,
            "humidity": 73,

```

```

    "temp_kf": 0
  },
  "weather": [
    {
      "id": 501,
      "main": "Rain",
      "description": "moderate rain",
      "icon": "10n"
    }
  ],
  "clouds": {
    "all": 100
  },
  "wind": {
    "speed": 6.62,
    "deg": 234,
    "gust": 11.02
  },
  "visibility": 8224,
  "pop": 1,
  "rain": {
    "3h": 6.51
  },
  "sys": {
    "pod": "n"
  },
  "dt_txt": "2024-08-01 15:00:00"
},
{
  "dt": 1722535200,
  "main": {
    "temp": 300.76,
    "feels_like": 304.17,
    "temp_min": 300.76,
    "temp_max": 300.76,
    "pressure": 1001,
    "sea_level": 1001,
    "grnd_level": 996,
    "humidity": 79,
    "temp_kf": 0
  },
  "weather": [
    {
      "id": 500,
      "main": "Rain",
      "description": "light rain",
      "icon": "10n"
    }
  ],
  "clouds": {
    "all": 100
  },
  "wind": {
    "speed": 6.86,
    "deg": 228,
    "gust": 11.66
  },
  "visibility": 10000,
  "pop": 1,
  "rain": {
    "3h": 0.65
  },
  "sys": {
    "pod": "n"
  },

```

```

    "dt_txt": "2024-08-01 18:00:00"
  },
  {
    "dt": 1722546000,
    "main": {
      "temp": 300.21,
      "feels_like": 303.12,
      "temp_min": 300.21,
      "temp_max": 300.21,
      "pressure": 1000,
      "sea_level": 1000,
      "grnd_level": 994,
      "humidity": 81,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 500,
        "main": "Rain",
        "description": "light rain",
        "icon": "10n"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 5.44,
      "deg": 221,
      "gust": 10.91
    },
    "visibility": 10000,
    "pop": 0.2,
    "rain": {
      "3h": 0.17
    },
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-08-01 21:00:00"
  },
  {
    "dt": 1722556800,
    "main": {
      "temp": 299.92,
      "feels_like": 302.55,
      "temp_min": 299.92,
      "temp_max": 299.92,
      "pressure": 1000,
      "sea_level": 1000,
      "grnd_level": 994,
      "humidity": 82,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 804,
        "main": "Clouds",
        "description": "overcast clouds",
        "icon": "04n"
      }
    ],
    "clouds": {
      "all": 100
    },
  },

```

```

    "wind": {
      "speed": 5.06,
      "deg": 221,
      "gust": 9.56
    },
    "visibility": 10000,
    "pop": 0,
    "sys": {
      "pod": "n"
    },
    "dt_txt": "2024-08-02 00:00:00"
  },
  {
    "dt": 1722567600,
    "main": {
      "temp": 301.05,
      "feels_like": 304.57,
      "temp_min": 301.05,
      "temp_max": 301.05,
      "pressure": 1001,
      "sea_level": 1001,
      "grnd_level": 996,
      "humidity": 77,
      "temp_kf": 0
    },
    "weather": [
      {
        "id": 500,
        "main": "Rain",
        "description": "light rain",
        "icon": "10d"
      }
    ],
    "clouds": {
      "all": 100
    },
    "wind": {
      "speed": 5.4,
      "deg": 233,
      "gust": 9
    },
    "visibility": 10000,
    "pop": 0.71,
    "rain": {
      "3h": 0.37
    },
    "sys": {
      "pod": "d"
    },
    "dt_txt": "2024-08-02 03:00:00"
  },
  {
    "dt": 1722578400,
    "main": {
      "temp": 303.4,
      "feels_like": 307.82,
      "temp_min": 303.4,
      "temp_max": 303.4,
      "pressure": 1001,
      "sea_level": 1001,
      "grnd_level": 995,
      "humidity": 66,
      "temp_kf": 0
    },
    "weather": [

```



```

        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 6.93,
        "deg": 255,
        "gust": 8.96
    },
    "visibility": 10000,
    "pop": 1,
    "rain": {
        "3h": 1.37
    },
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-08-02 06:00:00"
},
{
    "dt": 1722589200,
    "main": {
        "temp": 302.58,
        "feels_like": 306.63,
        "temp_min": 302.58,
        "temp_max": 302.58,
        "pressure": 999,
        "sea_level": 999,
        "grnd_level": 994,
        "humidity": 69,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10d"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 6.38,
        "deg": 248,
        "gust": 8.72
    },
    "visibility": 10000,
    "pop": 0.92,
    "rain": {
        "3h": 0.82
    },
    "sys": {
        "pod": "d"
    },
    "dt_txt": "2024-08-02 09:00:00"
},
{

```

```

"dt": 1722600000,
"main": {
  "temp": 301.29,
  "feels_like": 304.97,
  "temp_min": 301.29,
  "temp_max": 301.29,
  "pressure": 999,
  "sea_level": 999,
  "grnd_level": 994,
  "humidity": 76,
  "temp_kf": 0
},
"weather": [
  {
    "id": 500,
    "main": "Rain",
    "description": "light rain",
    "icon": "10d"
  }
],
"clouds": {
  "all": 100
},
"wind": {
  "speed": 4.82,
  "deg": 245,
  "gust": 8.51
},
"visibility": 10000,
"pop": 0.87,
"rain": {
  "3h": 0.28
},
"sys": {
  "pod": "d"
},
"dt_txt": "2024-08-02 12:00:00"
},
{
  "dt": 1722610800,
  "main": {
    "temp": 300.61,
    "feels_like": 303.94,
    "temp_min": 300.61,
    "temp_max": 300.61,
    "pressure": 1001,
    "sea_level": 1001,
    "grnd_level": 996,
    "humidity": 80,
    "temp_kf": 0
  },
  "weather": [
    {
      "id": 500,
      "main": "Rain",
      "description": "light rain",
      "icon": "10n"
    }
  ],
  "clouds": {
    "all": 100
  },
  "wind": {
    "speed": 4.39,
    "deg": 237,

```

```

        "gust": 8.65
      },
      "visibility": 10000,
      "pop": 0.95,
      "rain": {
        "3h": 0.48
      },
      "sys": {
        "pod": "n"
      },
      "dt_txt": "2024-08-02 15:00:00"
    },
    {
      "dt": 1722621600,
      "main": {
        "temp": 300.15,
        "feels_like": 303.27,
        "temp_min": 300.15,
        "temp_max": 300.15,
        "pressure": 1002,
        "sea_level": 1002,
        "grnd_level": 996,
        "humidity": 84,
        "temp_kf": 0
      },
      "weather": [
        {
          "id": 500,
          "main": "Rain",
          "description": "light rain",
          "icon": "10n"
        }
      ],
      "clouds": {
        "all": 100
      },
      "wind": {
        "speed": 3.9,
        "deg": 219,
        "gust": 7.78
      },
      "visibility": 10000,
      "pop": 1,
      "rain": {
        "3h": 0.73
      },
      "sys": {
        "pod": "n"
      },
      "dt_txt": "2024-08-02 18:00:00"
    },
    {
      "dt": 1722632400,
      "main": {
        "temp": 299.73,
        "feels_like": 299.73,
        "temp_min": 299.73,
        "temp_max": 299.73,
        "pressure": 1000,
        "sea_level": 1000,
        "grnd_level": 995,
        "humidity": 86,
        "temp_kf": 0
      },
      "weather": [

```

```

        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 4.29,
        "deg": 237,
        "gust": 9.58
    },
    "visibility": 10000,
    "pop": 1,
    "rain": {
        "3h": 1.92
    },
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2024-08-02 21:00:00"
},
{
    "dt": 1722643200,
    "main": {
        "temp": 299.55,
        "feels_like": 299.55,
        "temp_min": 299.55,
        "temp_max": 299.55,
        "pressure": 1000,
        "sea_level": 1000,
        "grnd_level": 995,
        "humidity": 87,
        "temp_kf": 0
    },
    "weather": [
        {
            "id": 500,
            "main": "Rain",
            "description": "light rain",
            "icon": "10n"
        }
    ],
    "clouds": {
        "all": 100
    },
    "wind": {
        "speed": 4.42,
        "deg": 237,
        "gust": 9.93
    },
    "visibility": 10000,
    "pop": 1,
    "rain": {
        "3h": 0.75
    },
    "sys": {
        "pod": "n"
    },
    "dt_txt": "2024-08-03 00:00:00"
},
{

```

```

        "dt": 1722654000,
        "main": {
            "temp": 301.13,
            "feels_like": 305.05,
            "temp_min": 301.13,
            "temp_max": 301.13,
            "pressure": 1002,
            "sea_level": 1002,
            "grnd_level": 997,
            "humidity": 79,
            "temp_kf": 0
        },
        "weather": [
            {
                "id": 500,
                "main": "Rain",
                "description": "light rain",
                "icon": "10d"
            }
        ],
        "clouds": {
            "all": 100
        },
        "wind": {
            "speed": 4.99,
            "deg": 247,
            "gust": 7.6
        },
        "visibility": 10000,
        "pop": 0.7,
        "rain": {
            "3h": 0.3
        },
        "sys": {
            "pod": "d"
        },
        "dt_txt": "2024-08-03 03:00:00"
    },
    ],
    "city": {
        "id": 1279233,
        "name": "Ahmedabad",
        "coord": {
            "lat": 23.0216,
            "lon": 72.5797
        },
        "country": "IN",
        "population": 3719710,
        "timezone": 19800,
        "sunrise": 1722213547,
        "sunset": 1722261178
    }
}

```

```

In [7]: #Fetch the below data from API
D = {"date_time":[], "temp":[], "pressure":[], "humidity":[], "weather":[], "wind_speed"

for i in five_day['list']:
    D["date_time"].append(i["dt_txt"])
    D["temp"].append(i["main"]["temp"])
    D["pressure"].append(i["main"]["pressure"])
    D["humidity"].append(i["main"]["humidity"])
    D["weather"].append(i["weather"][0]["description"])
    D["wind_speed"].append(i["wind"]["speed"])

```


	date_time	temp	pressure	humidity	weather	wind_speed
13	2024-07-30 21:00:00	300.75	1002	81	light rain	4.56
14	2024-07-31 00:00:00	300.14	1002	84	overcast clouds	4.05
15	2024-07-31 03:00:00	301.75	1003	75	overcast clouds	4.47
16	2024-07-31 06:00:00	305.97	1003	54	light rain	5.46
17	2024-07-31 09:00:00	306.53	1002	49	light rain	4.86
18	2024-07-31 12:00:00	305.32	1000	54	overcast clouds	3.42
19	2024-07-31 15:00:00	304.06	1002	61	overcast clouds	5.73
20	2024-07-31 18:00:00	302.43	1003	70	overcast clouds	4.99
21	2024-07-31 21:00:00	301.20	1001	77	overcast clouds	5.66
22	2024-08-01 00:00:00	300.57	1001	80	overcast clouds	4.99
23	2024-08-01 03:00:00	301.63	1003	73	overcast clouds	6.26
24	2024-08-01 06:00:00	303.74	1002	62	light rain	7.11
25	2024-08-01 09:00:00	305.85	1000	54	light rain	6.63
26	2024-08-01 12:00:00	304.09	999	63	moderate rain	5.64
27	2024-08-01 15:00:00	302.04	1001	73	moderate rain	6.62
28	2024-08-01 18:00:00	300.76	1001	79	light rain	6.86
29	2024-08-01 21:00:00	300.21	1000	81	light rain	5.44
30	2024-08-02 00:00:00	299.92	1000	82	overcast clouds	5.06
31	2024-08-02 03:00:00	301.05	1001	77	light rain	5.40
32	2024-08-02 06:00:00	303.40	1001	66	light rain	6.93
33	2024-08-02 09:00:00	302.58	999	69	light rain	6.38
34	2024-08-02 12:00:00	301.29	999	76	light rain	4.82
35	2024-08-02 15:00:00	300.61	1001	80	light rain	4.39
36	2024-08-02 18:00:00	300.15	1002	84	light rain	3.90
37	2024-08-02 21:00:00	299.73	1000	86	light rain	4.29
38	2024-08-03 00:00:00	299.55	1000	87	light rain	4.42
39	2024-08-03 03:00:00	301.13	1002	79	light rain	4.99

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